Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.344 Hz*
Freq. upper bound:	*6.875 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*23*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*28*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *38.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1274*	*0.04943*
Peak wave period T_P (s):	*1.455*	*1.391*
Energy wave period $T_{-1,0}$ (s):	*1.341*	*1.235*
Mean wave period $T_{0,1}$ (s):	*1.251*	*1.005*
Mean wave period $T_{0,2}$ (s):	*1.184*	*0.8777*
Spectral width (Broadness):	*0.7543*	*0.8266*
Spectral width (Narrowness):	*0.3397*	*0.5568*

Time Domain Analysis

Overall amplitude reflection coefficient: *38.9%*

	Incident	**Reflected**
Number of waves:	961	1184
Mean wave height H_m (m):	*0.08288*	*0.03181*
Mean wave period T_m (s):	*1.264*	*1.026*
Sig. wave height H_s (m):	*0.1261*	*0.04839*
$T_{H_{1/3}}$ (s):	*1.395*	*1.288*
H_{max} (m):	*0.2008*	*0.08644*
T_{Hmax} (s):	*1.436*	*1.298*
$H_{1/10}$ (m):	*0.1512*	*0.0594*
$H_{1/50}$ (m):	*0.1734*	*0.06859*
$H_{1/100}$ (m)	*0.1833*	*0.07172*
$H_{1/250}$ (m):	*0.1937*	*0.07698*
$H_{0.1\%}$ (m):	*Not enough data*	*0.08565*
$H_{1\%}$ (m):	*0.1711*	*0.06734*
$H_{2\%}$ (m):	*0.159*	*0.06397*
$H_{10\%}$ (m)	*0.1347*	*0.05199*
Groupiness factor GF:	*0.9229*	*0.9898*
Skewness b1:	*0.4205*	*0.1869*
Kurtosis b2:	*2.84*	*3.026*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *39.7%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.137*	*0.05446*
Peak wave period T_P (s):	*1.524*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.402*	*1.3*
Mean wave period $T_{0,1}$ (s):	*1.299*	*1.048*
Mean wave period $T_{0,2}$ (s):	*1.225*	*0.91*
Spectral width (Broadness):	*0.7619*	*0.8294*
Spectral width (Narrowness):	*0.3542*	*0.5718*

Time Domain Analysis

Overall amplitude reflection coefficient: *40%*

	Incident	Reflected
Number of waves:	975	1185
Mean wave height H_m (m):	*0.09027*	*0.03576*
Mean wave period T_m (s):	*1.328*	*1.093*
Sig. wave height H_s (m):	*0.136*	*0.05442*
$T_{H_{1/3}}$ (s):	*1.453*	*1.392*
H_{max} (m):	*0.1925*	*0.0938*
T_{Hmax} (s):	*1.552*	*1.626*
$H_{1/10}$ (m):	*0.1607*	*0.06562*
$H_{1/50}$ (m):	*0.1817*	*0.07624*
$H_{1/100}$ (m)	*0.1866*	*0.08064*
$H_{1/250}$ (m):	*0.1909*	*0.08719*
$H_{0.1\%}$ (m):	*Not enough data*	*0.09347*
$H_{1\%}$ (m):	*0.1802*	*0.07436*
$H_{2\%}$ (m):	*0.1741*	*0.07001*
$H_{10\%}$ (m)	*0.1454*	*0.05865*
Groupiness factor GF:	*0.9086*	*0.9926*
Skewness b1:	*0.466*	*0.2587*
Kurtosis b2:	*2.794*	*2.97*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*26*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *41.4%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1465*	*0.06063*
Peak wave period T_P (s):	*1.524*	*1.561*
Energy wave period $T_{-1,0}$ (s):	*1.449*	*1.364*
Mean wave period $T_{0,1}$ (s):	*1.33*	*1.082*
Mean wave period $T_{0,2}$ (s):	*1.244*	*0.9272*
Spectral width (Broadness):	*0.7873*	*0.8431*
Spectral width (Narrowness):	*0.3783*	*0.6007*

Time Domain Analysis

Overall amplitude reflection coefficient: *41.4%*

	Incident	Reflected
Number of waves:	990	1188
Mean wave height H_m (m):	*0.09643*	*0.03939*
Mean wave period T_m (s):	*1.373*	*1.144*
Sig. wave height H_s (m):	*0.1441*	*0.05981*
$T_{H_{1/3}}$ (s):	*1.519*	*1.482*
H_{max} (m):	*0.1978*	*0.1081*
T_{Hmax} (s):	*1.745*	*1.54*
$H_{1/10}$ (m):	*0.1684*	*0.07262*
$H_{1/50}$ (m):	*0.1862*	*0.08574*
$H_{1/100}$ (m)	*0.1918*	*0.09042*
$H_{1/250}$ (m):	*0.1943*	*0.09777*
$H_{0.1\%}$ (m):	*Not enough data*	*0.1068*
$H_{1\%}$ (m):	*0.1883*	*0.08305*
$H_{2\%}$ (m):	*0.1784*	*0.07901*
$H_{10\%}$ (m)	*0.1529*	*0.06332*
Groupiness factor GF:	*0.8859*	*0.9792*
Skewness b1:	*0.5182*	*0.282*
Kurtosis b2:	*2.762*	*2.956*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.313 Hz*
Freq. upper bound:	*6.250 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*31*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *42.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1573*	*0.06729*
Peak wave period T_P (s):	*1.6*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.497*	*1.437*
Mean wave period $T_{0,1}$ (s):	*1.36*	*1.115*
Mean wave period $T_{0,2}$ (s):	*1.261*	*0.9416*
Spectral width (Broadness):	*0.806*	*0.8538*
Spectral width (Narrowness):	*0.403*	*0.6339*

Time Domain Analysis

Overall amplitude reflection coefficient: *42.8%*

	Incident	**Reflected**
Number of waves:	975	1157
Mean wave height H_m (m):	*0.1047*	*0.04442*
Mean wave period T_m (s):	*1.435*	*1.209*
Sig. wave height H_s (m):	*0.1535*	*0.0668*
$T_{H_{1/3}}$ (s):	*1.594*	*1.561*
H_{max} (m):	*0.2292*	*0.104*
T_{Hmax} (s):	*1.481*	*1.512*
$H_{1/10}$ (m):	*0.1805*	*0.0802*
$H_{1/50}$ (m):	*0.2074*	*0.09195*
$H_{1/100}$ (m)	*0.2154*	*0.09623*
$H_{1/250}$ (m):	*0.224*	*0.1004*
$H_{0.1\%}$ (m):	*Not enough data*	*0.1034*
$H_{1\%}$ (m):	*0.2054*	*0.09113*
$H_{2\%}$ (m):	*0.1956*	*0.08567*
$H_{10\%}$ (m)	*0.1628*	*0.07084*
Groupiness factor GF:	*0.8703*	*0.9422*
Skewness b1:	*0.5635*	*0.2308*
Kurtosis b2:	*2.732*	*2.848*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.305 Hz*
Freq. upper bound:	*6.094 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*32*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *44.1%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1641*	*0.07237*
Peak wave period T_P (s):	*1.641*	*1.778*
Energy wave period $T_{-1,0}$ (s):	*1.539*	*1.512*
Mean wave period $T_{0,1}$ (s):	*1.387*	*1.158*
Mean wave period $T_{0,2}$ (s):	*1.28*	*0.9708*
Spectral width (Broadness):	*0.8116*	*0.8564*
Spectral width (Narrowness):	*0.4174*	*0.651*

Time Domain Analysis

Overall amplitude reflection coefficient: *44.2%*

	Incident	Reflected
Number of waves:	1011	1187
Mean wave height H_m (m):	*0.1103*	*0.04822*
Mean wave period T_m (s):	*1.463*	*1.246*
Sig. wave height H_s (m):	*0.1598*	*0.07196*
$T_{H_{1/3}}$ (s):	*1.639*	*1.626*
H_{max} (m):	*0.2134*	*0.1174*
T_{Hmax} (s):	*1.628*	*1.518*
$H_{1/10}$ (m):	*0.1839*	*0.08612*
$H_{1/50}$ (m):	*0.2016*	*0.09908*
$H_{1/100}$ (m)	*0.2063*	*0.1043*
$H_{1/250}$ (m):	*0.21*	*0.1125*
$H_{0.1\%}$ (m):	*0.2134*	*0.1166*
$H_{1\%}$ (m):	*0.202*	*0.09645*
$H_{2\%}$ (m):	*0.1943*	*0.09195*
$H_{10\%}$ (m)	*0.1699*	*0.07605*
Groupiness factor GF:	*0.8474*	*0.9107*
Skewness b1:	*0.5924*	*0.2267*
Kurtosis b2:	*2.666*	*2.797*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *45.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1673*	*0.07628*
Peak wave period T_P (s):	*1.73*	*1.939*
Energy wave period $T_{-1,0}$ (s):	*1.582*	*1.585*
Mean wave period $T_{0,1}$ (s):	*1.408*	*1.186*
Mean wave period $T_{0,2}$ (s):	*1.29*	*0.9806*
Spectral width (Broadness):	*0.8214*	*0.8613*
Spectral width (Narrowness):	*0.4383*	*0.6805*

Time Domain Analysis

Overall amplitude reflection coefficient: *45.6%*

	Incident	**Reflected**
Number of waves:	1157	1351
Mean wave height H_m (m):	*0.1156*	*0.05189*
Mean wave period T_m (s):	*1.534*	*1.314*
Sig. wave height H_s (m):	*0.1639*	*0.0768*
$T_{H_{1/3}}$ (s):	*1.668*	*1.723*
H_{max} (m):	*0.2251*	*0.1351*
T_{Hmax} (s):	*1.653*	*1.394*
$H_{1/10}$ (m):	*0.186*	*0.09234*
$H_{1/50}$ (m):	*0.207*	*0.1069*
$H_{1/100}$ (m)	*0.2154*	*0.1122*
$H_{1/250}$ (m):	*0.2216*	*0.1223*
$H_{0.1\%}$ (m):	*0.2247*	*0.1334*
$H_{1\%}$ (m):	*0.2048*	*0.103*
$H_{2\%}$ (m):	*0.1951*	*0.1001*
$H_{10\%}$ (m)	*0.1707*	*0.08113*
Groupiness factor GF:	*0.8386*	*0.9234*
Skewness b1:	*0.6079*	*0.1888*
Kurtosis b2:	*2.659*	*2.849*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.dat (4 cols)*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.453 Hz*
Freq. upper bound:	*9.063 Hz*

Freq. Domain Analysis

Overlap (FFT blocks):	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*20*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*21*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *30.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.06823*	*0.02096*
Peak wave period T_P (s):	*1.103*	*1.103*
Energy wave period $T_{-1,0}$ (s):	*1.057*	*0.9789*
Mean wave period $T_{0,1}$ (s):	*1.024*	*0.8605*
Mean wave period $T_{0,2}$ (s):	*1.003*	*0.7902*
Spectral width (Broadness):	*0.4974*	*0.7299*
Spectral width (Narrowness):	*0.2047*	*0.4311*

Time Domain Analysis

Overall amplitude reflection coefficient: *30.7%*

	Incident	**Reflected**
Number of waves:	1016	1204
Mean wave height H_m (m):	*0.04284*	*0.01323*
Mean wave period T_m (s):	*1.015*	*0.8569*
Sig. wave height H_s (m):	*0.06719*	*0.02034*
$T_{H_{1/3}}$ (s):	*1.05*	*1.027*
H_{max} (m):	*0.118*	*0.0344*
T_{Hmax} (s):	*1.034*	*1.035*
$H_{1/10}$ (m):	*0.08474*	*0.02509*
$H_{1/50}$ (m):	*0.1014*	*0.03009*
$H_{1/100}$ (m)	*0.1071*	*0.03179*
$H_{1/250}$ (m):	*0.1128*	*0.03324*
$H_{0.1\%}$ (m):	*0.1179*	*0.03428*
$H_{1\%}$ (m):	*0.09962*	*0.03011*
$H_{2\%}$ (m):	*0.09281*	*0.02684*
$H_{10\%}$ (m)	*0.07239*	*0.02156*
Groupiness factor GF:	*0.979*	*0.9816*
Skewness b1:	*0.02253*	*0.01533*
Kurtosis b2:	*2.959*	*2.948*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-23 reference-case- 4% - test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.398 Hz*
Freq. upper bound:	*7.969 Hz*

Freq. Domain Analysis

Overlap (FFT) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*22*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*24*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *35.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1049*	*0.03731*
Peak wave period T_P (s):	*1.255*	*1.28*
Energy wave period $T_{-1,0}$ (s):	*1.217*	*1.074*
Mean wave period $T_{0,1}$ (s):	*1.161*	*0.9024*
Mean wave period $T_{0,2}$ (s):	*1.12*	*0.8105*
Spectral width (Broadness):	*0.6682*	*0.7755*
Spectral width (Narrowness):	*0.2723*	*0.4894*

Time Domain Analysis

Overall amplitude reflection coefficient: *35.6%*

	Incident	**Reflected**
Number of waves:	1003	1296
Mean wave height H_m (m):	*0.06765*	*0.02363*
Mean wave period T_m (s):	*1.156*	*0.894*
Sig. wave height H_s (m):	*0.1034*	*0.03519*
$T_{H_{1/3}}$ (s):	*1.243*	*1.075*
H_{max} (m):	*0.1578*	*0.06859*
T_{Hmax} (s):	*1.222*	*1.402*
$H_{1/10}$ (m):	*0.1265*	*0.04343*
$H_{1/50}$ (m):	*0.1459*	*0.05218*
$H_{1/100}$ (m)	*0.1515*	*0.05534*
$H_{1/250}$ (m):	*0.1553*	*0.0591*
$H_{0.1\%}$ (m):	*0.1578*	*0.06551*
$H_{1\%}$ (m):	*0.1448*	*0.05108*
$H_{2\%}$ (m):	*0.1364*	*0.04743*
$H_{10\%}$ (m)	*0.11*	*0.03772*
Groupiness factor GF:	*0.9278*	*0.9897*
Skewness b1:	*0.1548*	*-0.03401*
Kurtosis b2:	*2.86*	*2.994*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*23. mrt 2026*
Remarks	
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